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21 APRIL 1952
Superseding
MIL-T-6342(USAF)
8 January 1951

MILITARY SPECIFICATION

TEST SET, SLAVED GYRO MAGNETIC COMPASS, FIELD, TYPE P-1

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of slaved gyro magnetic compass test set, designated type P-1, for use in determining the serviceability of slaved gyro magnetic compasses in use in the service.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Military

MIL-E-5272(USAF)	Environmental Testing, Aeronautical and Associated Equipment, (General Specification For)
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers

2.1.2 Other Publications.-

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

2.1.3 Drawings.-

U. S. Air Force Drawing

47R7466	Tester Assembly - Slaved Gyro Magnetic Compass, Field, Type P-1

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The requirements specified in Drawing 47R7466 are applicable as requirements for this specification.

3.1.1 In the event the requirements of Drawing 47R7466 and this specification conflict, the requirements of this specification shall govern.

3.2 Materials.-

3.2.1 Protective Treatment.- When materials are used in the construction of the test set that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.2.2 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with AMA Bulletin No. 143, except as provided in the following paragraph.

3.2.2.1 AN or JAN Standard Parts.- AN or JAN Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or JAN Standard parts without alteration, and provided the corresponding AN or JAN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design.- The test set shall be designed to indicate the serviceability of slaved gyro magnetic compasses.

3.4 Construction.- The test set shall be so constructed that no parts will work loose in service. The test set shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 The test set shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases.

3.5 Identification of Product.-

3.5.1 Nameplate.- A nameplate, permanently and legibly filled in with the information listed below, shall be securely attached to the test set. The information marked in the spaces provided shall be in accordance with Specification MIL-M-7911.

TEST SET, SLAVED GYRO MAGNETIC COMPASS, FIELD
Type P-1
Specification MIL-T-6342A
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

3.6 Workmanship.- The test set, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding, and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required for the testing of test sets are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall make available test reports for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as guaranty of the acceptance of the finished product.

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4.2 Test Conditions.-

4.2.1 Standard Atmospheric Conditions.- Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.2.2 Attitude.- Unless otherwise specified, the test set shall be tested in normal operating position, that is, with the panel horizontal.

4.2.3 Tapping.- Unless otherwise specified, the test set shall be gently tapped or vibrated before a test reading is taken.

4.3 Individual Tests.- Each test set shall be subjected to the following tests.

4.3.1 Examination of Product.- Each test set shall be inspected to determine compliance with this specification with respect to material and workmanship.

4.3.2 Functional.- Each completed test set shall be subjected to a functional test to determine that component parts are properly assembled, to check continuity of wiring, and to insure that the test set operates properly. The functional test shall be accomplished on the meter panel as shown on figure 1.

4.3.2.1 Alternating Current Voltage.- Apply 115 ±2V; 400 ±4 cycles; one-phase ac to pins "A - B" on the four-pin panel receptacle. With Switch No. 1 on "Power check" and Switch No. 5 on "A - B," the pointer of the meter shall deflect to the 115V indication ±3 volts. Apply the same voltage to pins "B - C" and "A - C" respectively. The pointer of the meter shall defect to the 115V indication ±3 volts.

4.3.2.2 Direct Current Voltage.- Apply 30 ±1V dc to pins "A - D" on the four-pin panel receptable. With Switch No. 1 on "power check" and Switch No. 5 on "28V dc," the pointer of the meter shall deflect to the 30V indication ±3 volts.

4.3.2.3 Frequency.- During the test described in paragraph covering Alternating current voltage, observe the indication of the frequency meter which shall indicate correctly within ±4 cycles.

4.3.2.4 Phase Rotation.- Apply 115 ±5V, 400 ±20 cycles, three-phase ac to pins "A - B - C" on the four-pin panel receptacle. When the phase rotation of the input current is "A - B - C," the green panel light shall operate. When the phase rotation of the input current is "C - B - A," the red panel light shall operate.

4.3.2.5 Ground Check.- Apply 115 ±5V, 400 ±20 cycles, three-phase ac to pins "A - B - C," on the four-pin receptacle. When the phase rotation has been determined to be correct from the above test, set Switch No. 1 to "Ground check." Connect a lead from the banana jack on the instrument panel to electrical ground. If any phase other than "A" phase is at ground potential, the green panel light shall operate. Disregard operation of red light during this test.

4.3.2.6 Circuit Continuity.-

4.3.2.6.1 Flux Valve Selector Circuit.- Apply 115V, 400-cycles, single-phase ac to pins "A - C" on the four-pin panel receptacle. With Switch No. 1 on "Flux valve check," and Switch No. 3 on "Calibrate," rotate rheostat No. 4 together with adjustable stop until the pointer of the meter deflects to the line marked with ADJ. Return rheostat No. 4 to "OFF" position and set Switch No. 3 on "Exciter." Connect a noninductive resistance decade box, General Radio Type 602J or equal, across pins "I - J" on the 10-pin receptacle. Set the decade box to 1,475 ohms and turn rheostat No. 4 until it touches the adjustable stop. The pointer of the meter shall deflect to cross-hatched boundary "E," as referenced on figure 1, within 1/32 inch. Return rheostat No. 4 to the "OFF" position, and reset the decade box to 610 ohms. Check this value on the meter, as above, for boundary "F." Return rheostat to "OFF" position and change decade box from pins "I - J" to pins "F - J." Set in a value of 52 ohms on the decade box. Set Switch No. 3 in "A" position and turn rheostat

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No. 4 until it touches the adjustable stop. The pointer of the meter shall deflect to boundary "G," as referenced on figure 1, within ±1/32 inch. Repeat this operation with Switch No. 3 set on "B" position and the decade box set at 52 ohms and connected between pins "G - J." The same prccedure shall be repeated with Switch No. 3 on "C" position and the decade box set at 52 ohms and connected between pins "H - J." Set decade box to 20 ohms and repeat the test with Switch No. 3 in position "A" as above. The pointer of the meter shall deflect to the boundary "H," as referenced on figure 1, within ±1/32 inch.

4.3.2.6.2 Slave Check Selector Circuit.- Set Switch No. 1 on the Slave check position. Then with Switch No. 2 set in the various positions as shown in column 1 of table I conduct test to insure continuity of circuits between pins on the 10-pin panel receptacle as shown in column 2 of table I.

TABLE I

Column 1	Column 2
Position of Switch No. 2	Closed Circuit between Pins
"O"	A-F
"O"	B-G
"O"	C-H
"O"	D-I
"O"	E-J
"R"	A-H
"R"	B-F
"R"	C-G
"R"	D-I
"R"	E-J
"L"	A-G
"L"	B-H
"L"	C-F
"L"	D-I
"L"	E-J

4.3.2.6.3 Switch Interlock.- When Rheostat No. 4 is set in the "OFF" position, Switches No. 3 and 1 shall operate freely. When Rheostat No. 4 is rotated from the "OFF" position, Switches No. 3 and 1 shall be interlocked and impossible to turn. When Switch No. 1 is set in either the "Slave check" or "Power check" position, Switch No. 3 should move freely, and Rheostat No. 4 shall be interlocked in the "OFF" position.

4.4 Sampling Tests.- One test set shall be selected at random from each lot of 100 or fraction thereof on the order and subjected to the following tests. These tests shall be in addition to the Individual tests. A lot shall consist of test sets manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.1 Low Temperature.- The equipment shall be placed within a test chamber capable of maintaining an internal temperature of -54° ±2°C for a period of 24 hours. At the end of the 24-hour period the temperature shall be changed to -40° ±2°C. At the end of an additional 2-hour period the equipment shall meet the Individual test requirements specified herein. There shall be no damage which would affect subsequent operation.

4.4.2 High Temperature.- The equipment shall be placed within a test chamber capable of maintaining an internal temperature of +71° ±2°C for a period of 24 hours. At the end of the 24-hour period, the temperature shall be changed to +55° ±2°C. At the end of an additional 2-hour period the equipment shall meet the Individual test requirements specified herein. There shall be no damage which would adversely affect subsequent operation.

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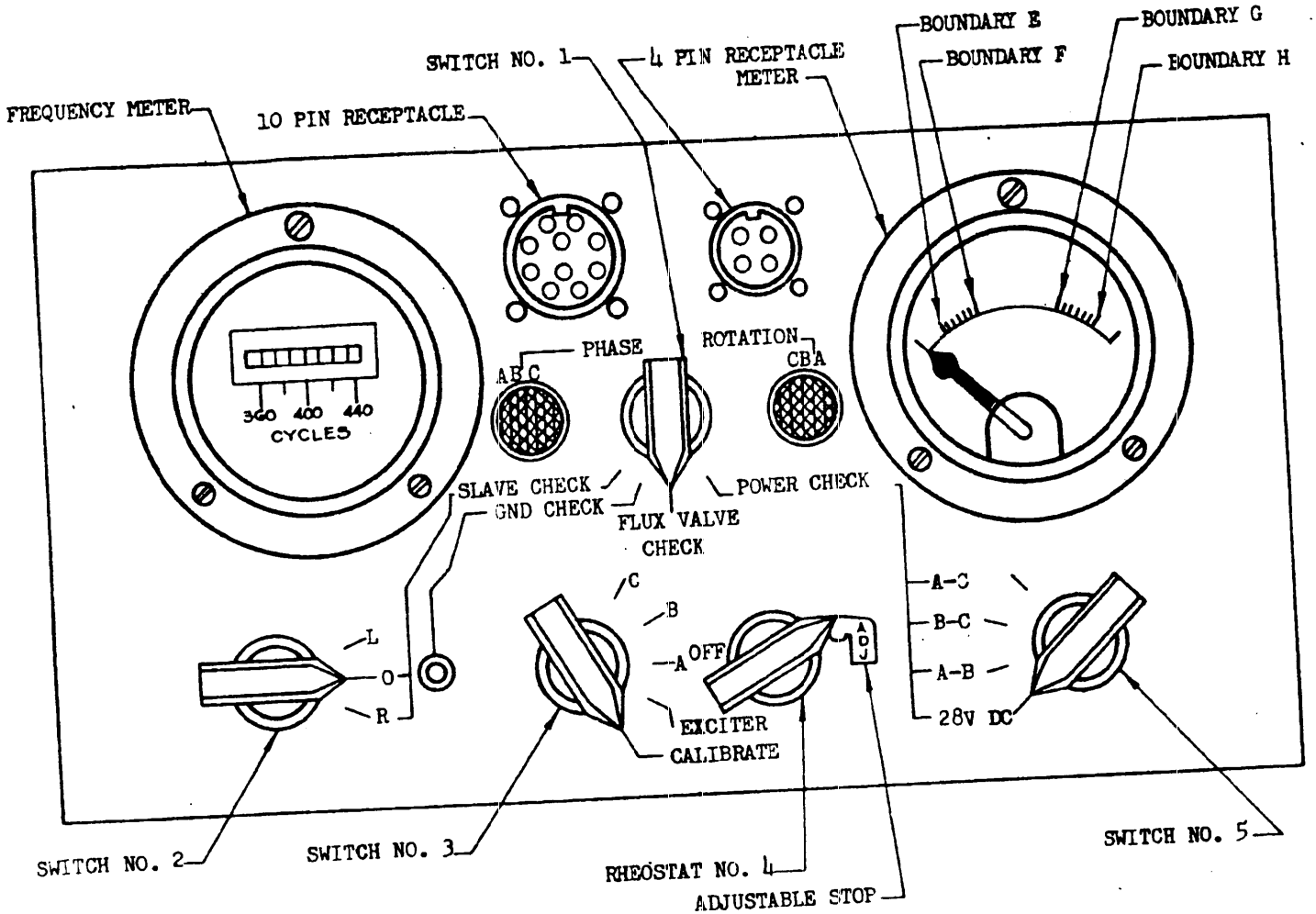


FIGURE 1. Panel Layout

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4.4.3 Fungus Resistance.- The fungus resistance test shall be in accordance with Fungus Resistance test, Procedure I of Specification MIL-E-5272(USAF), except that the test period shall be 14 days. At the conclusion of the test period the test set shall be carefully examined to ascertain that no fungus growth has occurred. In addition there shall be no other damage which would adversely affect subsequent operation.

4.5 Rejection and Retest.-

4.5.1 Items which have been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.5.2 When tests are specified on a quantity of test sets that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, one additional test set of the lot represented shall be tested immediately to determine the extent of failure. If one or more test sets of the additional samples fail to meet the specified tests, the entire lot shall be rejected. Individual performance tests need not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the tester.

4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification and drawings, to determine compliance with the specification and drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and Packing.- Each test set shall be packaged and packed for shipment in accordance with Specification MIL-P-5633.

5.3 Marking and Labeling.- Marking shall be as specified in Specification MIL-P-5633. In addition each container shall be marked with the following information:

TEST SET, SLAVED GYRO MAGNETIC COMPASS, FIELD
Type P-1
Specification MIL-T-6342A
Drawing No. 47R7466
Stock No.
Date of Manufacture
Quantity
Contract or Order No.
Manufacturer's Name or Trade-Mark
Name of Contractor (if different from the manufacturer)

IF IN STORAGE AFTER DATE*, THE TEST SET SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

* This date shall be 18 months from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

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5.3.1 Precautionary Marking.- The following precautionary marking shall be included on the exterior shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Ordering Data.- Procurement documents should specify the title, number, and date of this specification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force

Other interest:
Navy - A

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